

QUALITY360 NEXT

STEP 4 PRD

PRODUCT QUALITY TESTING

Version: 1.0

Target Customer: Vedanta Aluminium – Jharsuguda

Platform: Quality360

Depends On:

Step 1 – Incoming Material Inspection

Step 2 – Process Material Qualification

Step 3 – Metal Quality Control

Quality Genealogy Framework

1. BUSINESS OBJECTIVE

Verify that finished aluminum products meet quality requirements before release for certification and dispatch.

This module represents the final product quality gate.

2. BUSINESS CONTEXT

After casting operations complete, finished products are generated.

Examples:

Primary Aluminum Ingots

Primary Aluminum Billets

The products must be inspected and tested before release.

3. BUSINESS QUESTION

Can this product batch be approved for customer release?

4. PRIMARY USERS

Production Engineer

Lab Analyst

QA Engineer

QA Manager

Viewer

5. BUSINESS FLOW

Metal Batch Released ↓ Product Batch Created ↓ Sample Collected ↓ Product Testing ↓ Validation ↓
Product Compliance Evaluation ↓ QA Review ↓ Product Approval

6. PRODUCT BATCH

Entity:

PB-2026-000210

Linked To:

Metal Batch

MB-2026-000789

Mandatory Genealogy Link

7. PAGE STRUCTURE

Product Batch Queue

Product Quality Workbench

Reuse existing framework.

8. PRODUCT BATCH QUEUE

Columns

Product Batch

Product Type

Weight

Status

Compliance

Created Date

Actions

Create

Open

Clone

Cancel

9. PRODUCT TYPES

Phase 1

Primary Aluminum Ingot

Primary Aluminum Billet

Future

Wire Rod

Alloy Billet

Value Added Products

10. WORKBENCH SECTIONS

Header

Workflow Timeline

Product Overview

Sample Management

Test Workspace

Quality Insights

Activity Feed

Approval Center

Audit Trail

Genealogy Card

11. TESTS

Mechanical Testing

UTS

Yield Strength

Elongation

Physical Testing

Hardness

Conductivity

Dimensions

Weight

Metallography

Microstructure Review

Surface Defects

Visual Inspection

12. INSTRUMENTS

UTS Machine

Hardness Tester

Metallurgical Microscope

Conductivity Analyzer

13. PRODUCT COMPLIANCE SCORE

Hero KPI

Product Compliance

0-100

Example

97/100

14. QUALITY INSIGHTS

Display

Product Compliance

Risk Level

Release Recommendation

Observations

Historical Comparison

15. RELEASE RECOMMENDATION

Outputs

Approve Product

Hold Product

Reject Product

Retest Product

16. OBSERVATIONS

Examples

Mechanical properties compliant.

Surface quality acceptable.

No significant defects observed.

Product suitable for customer release.

17. GENEALOGY

Display

RM Lot



Process Batch



Metal Batch



Product Batch

Current Node Highlighted

18. APPROVALS

Approve Product

Hold Product

Reject Product

Override

19. MOCK DEMO SCENARIO

Product Batch

PB-2026-000210

Product

Primary Aluminum Ingot

UTS

165 MPa

Hardness

52 HB

Conductivity

61 %

Compliance

97/100

Recommendation

APPROVE PRODUCT

SUCCESS CRITERIA

User can evaluate finished product quality and release product for certification.